Kali Linux Beginner Commands

50 simple slides for first-time Linux students

```
$ pwd
/home/student/kali-lab
```

Goal: understand commands by using one connected lab example.

What is Linux?

A simple operating system idea.

> Linux is an operating system, like Windows or macOS.

> It helps the computer run programs and manage files.

> Many servers, phones, labs, and security tools use Linux.

Remember Linux gives you strong control through the terminal.

What is Kali Linux?

A Linux system made for cybersecurity learning.

> Kali Linux is based on Debian Linux.

> It includes many security and networking tools.

> Use it only in your own lab or with clear permission.

Ethics Learning Kali is about responsible practice, not attacking real systems.

Why learn commands?

The terminal is fast, clear, and repeatable.

> Commands help you move, create, search, and fix files.

> They work well on servers and virtual machines.

> Short commands can do big tasks step by step.

Terminal idea

```
$ command options target
```

Example: `ls -l Documents`

Remember Do not memorize everything today. Understand the pattern.

Today we follow one simple lab

Every command helps build or inspect the same small project.

1. Where am I?
pwd, ls, cd

→

2. Build files
mkdir, touch, echo, cat

→

3. Organize
cp, mv, rm

→

4. Search
grep, find

→

5. System + network
ip, ping, curl, nmap

Class flow Students can copy each command exactly, then change names after they understand it.

pwd

Print Working Directory

COMMAND pwd

- Shows the folder you are in now.
- Use it when you feel lost.
- The path starts from the root folder /.

RUN THIS

```
$ pwd
/home/student
```

Lab tip Before creating files, check where you are.

ls

List files and folders

COMMAND `ls`

- Shows what is inside the current folder.
- Use -l for details.
- Use -a to see hidden files.

RUN THIS

```
$ ls
$ ls -la
```

```
Desktop  Documents  Downloads
```

Lab tip We will use ls after most file commands to check the result.

cd

Change Directory

COMMAND `cd`

> Moves you into another folder.

> Use .. to go one level back.

> Use ~ to go to your home folder.

RUN THIS

```
$ cd Documents
$ cd ..
$ cd ~
```

/home/student

Lab tip Think of cd as walking through folders.

mkdir

Make Directory

COMMAND mkdir

> Creates a new folder.

> Good for organizing a lab.

> Folder names should be short and clear.

RUN THIS

```
$ mkdir kali-lab
$ ls
```

Desktop Documents Downloads kali-lab

Lab tip Our main class folder is kali-lab.

touch

Create an empty file

COMMAND touch

> Creates a blank file quickly.

> Also updates the file time.

> Useful before editing notes.

RUN THIS

```
$ cd kali-lab
$ touch notes.txt
$ ls
```

```
notes.txt
```

Lab tip Now the lab has a notes file.

echo

Print text or write simple text

COMMAND echo

- Displays text in the terminal.
- Can send text into a file with >.
- Use >> to add more text later.

RUN THIS

```
$ echo "Kali class notes" >
notes.txt
$ echo "Command: pwd" >>
notes.txt
```

Lab tip > replaces file content. >> adds to the end.

cat

Show file content

COMMAND `cat`

- Prints a small file on the screen.
- Good for quick checking.
- Not ideal for very long files.

RUN THIS

```
$ cat notes.txt
```

```
Kali class notes
Command: pwd
```

Lab tip Use cat to confirm what echo wrote.

nano

Edit a text file

COMMAND nano

> Opens a simple terminal text editor.

> Good for beginner editing.

> Save with Ctrl+O, exit with Ctrl+X.

RUN THIS

```
$ nano notes.txt
```

Add: Command: ls

Lab tip Nano is easier for beginners than many advanced editors.

cp

Copy files or folders

COMMAND `cp`

- Makes a second copy.
- Original file stays in place.
- Use -r to copy folders.

RUN THIS

```
$ cp notes.txt backup-notes.txt
$ ls
```

```
backup-notes.txt  notes.txt
```

Lab tip Make backups before experimenting.

mv

Move or rename

COMMAND mv

> Moves a file to another place.

> Also renames files.

> No duplicate is created.

RUN THIS

```
$ mv backup-notes.txt old-notes.txt
$ ls
```

```
notes.txt  old-notes.txt
```

Lab tip mv is both move and rename.

rm

Remove files

COMMAND rm

- Deletes a file.
- Be careful: Linux may not ask again.
- Use rm only when you are sure.

RUN THIS

```
$ rm old-notes.txt
$ ls
```

```
notes.txt
```

Danger command: check pwd and ls first.

Careful Avoid rm -rf until students fully understand paths.

rmdir

Remove an empty folder

COMMAND **rmdir**

> Deletes a folder only if it is empty.

> Safer than rm -r for beginners.

> Useful for cleaning practice folders.

RUN THIS

```
$ mkdir temp
$ rmdir temp
$ ls
```

```
notes.txt
```

Lab tip If the folder has files, rmdir will refuse.

clear

Clean the terminal screen

COMMAND `clear`

> Hides old terminal output.

> Makes your screen easier to read.

> Does not delete files or commands.

RUN THIS

```
$ clear
```

Lab tip Use clear before explaining a new command.

history

Show previous commands

COMMAND history

> Lists commands you typed before.

> Helpful for review.

> Use !number to run a command again.

RUN THIS

```
$ history
```

```
1 pwd
2 mkdir kali-lab
3 cd kali-lab
```

Lab tip Students can use history to remember class steps.

man

Manual page

COMMAND `man`

> Shows the built-in manual for a command.

> Use q to quit.

> Best when you need full details.

RUN THIS

```
$ man ls
```

```
LS(1)  User Commands  LS(1)
```

Lab tip Manual pages look big, but the examples and options are useful.

--help

Quick command help

COMMAND `--help`

- Shows short help for many commands.
- Faster than man for beginners.
- Usually lists common options.

RUN THIS

```
$ ls --help
```

```
Usage: ls [OPTION]... [FILE]...
```

Lab tip Try command --help before searching online.

whoami

Show current user

COMMAND whoami

> Prints the username you are using.

> Useful before using sudo.

> Helps avoid permission confusion.

RUN THIS

```
$ whoami

student
```

Lab tip If it says root, be extra careful.

id

Show user and group IDs

COMMAND `id`

> Shows your user ID and groups.

> Groups affect permissions.

> Useful in multi-user Linux systems.

RUN THIS

```
$ id
```

```
uid=1000(student) gid=1000(student)
groups=1000(student),27(sudo)
```

Lab tip The sudo group means the user can run admin commands.

uname

Show system information

COMMAND `uname`

> Shows kernel information.

> Use -a for more detail.

> Good for checking the Linux system.

RUN THIS

```
$ uname -a
```

```
Linux kali 6.x-amd64 ... x86_64 GNU/Linux
```

Lab tip — Kernel means the core part of the operating system.

hostname

Show computer name

COMMAND hostname

> Prints the machine name.

> Helpful in labs with many computers.

> Names make systems easier to identify.

RUN THIS

```
$ hostname
```

```
kali
```

Lab tip A hostname is like a device nickname on a network.

date

Show current date and time

COMMAND `date`

- Displays system time.
- Useful for logs and screenshots.
- Can show if a VM clock is wrong.

RUN THIS

```
$ date
```

```
Sun Jul 5 19:00:00 +03 2026
```

Lab tip Correct time helps security tools and certificates work.

df

Check disk free space

COMMAND **df**

> Shows storage usage by filesystem.

> Use -h for human-readable sizes.

> Good before downloading big files.

RUN THIS

```
$ df -h
```

Filesystem	Size	Used	Avail	Use%
/dev/sda1	40G	15G	23G	40%

Lab tip Use df -h when a VM says disk is full.

du

Check folder size

COMMAND `du`

- Shows how much space files use.
- Use -sh for one simple total.
- Helpful for finding large folders.

RUN THIS

```
$ du -sh kali-lab
```

```
8.0K    kali-lab
```

Lab tip df checks disk space. du checks file or folder size.

free

Check memory usage

COMMAND `free`

> Shows RAM usage.

> Use -h for readable sizes.

> Helpful when the VM feels slow.

RUN THIS

```
$ free -h
```

```
Mem: 3.8Gi total 1.2Gi used 2.1Gi free
```

Lab tip RAM problems can make Kali slow inside a virtual machine.

top

Live process monitor

COMMAND top

> Shows running programs live.

> CPU and memory change in real time.

> Press q to quit.

RUN THIS

```
$ top
```

```
PID USER   %CPU %MEM COMMAND
1234 student 2.0  1.1 firefox
```

Lab tip Use top when the system becomes slow.

ps

List running processes

COMMAND `ps`

- Shows processes at one moment.
- Use aux for many details.
- Useful with grep to find a program.

RUN THIS

```
$ ps aux | grep nano
```

```
student 2345 0.0 0.1 nano notes.txt
```

Lab tip A process is a running program.

kill

Stop a process

COMMAND kill

> Stops a program by process ID.

> Use it only when a program is stuck.

> Find the PID with ps or top first.

RUN THIS

```
$ kill 2345
```

Careful Do not kill random system processes.

file

Detect file type

COMMAND `file`

- Tells what kind of file something is.
- Works even if the extension is wrong.
- Useful in security labs.

RUN THIS

```
$ file notes.txt
```

```
notes.txt: ASCII text
```

Lab tip Linux does not trust file extensions as much as Windows.

wc

Count lines, words, and bytes

COMMAND wc

- > Counts text information.
- > Use -l for lines.
- > Good for quick file checks.

RUN THIS

```
$ wc -l notes.txt
```

```
2 notes.txt
```

Lab tip wc means word count.

head

Show the first lines

COMMAND head

> Shows the top of a file.

> Default is 10 lines.

> Use -n to choose a number.

RUN THIS

```
$ head -n 2 notes.txt
```

```
Kali class notes
Command: pwd
```

Lab tip Use head for a quick first look.

tail

Show the last lines

COMMAND `tail`

- Shows the end of a file.
- Useful for logs.
- Use -f to follow new lines live.

RUN THIS

```
$ tail -n 2 notes.txt
```

```
Command: pwd
Command: ls
```

Lab tip tail -f is common when watching log files.

less

Read long files page by page

COMMAND `less`

> Opens a file in a scrollable viewer.

> Use arrows to move.

> Press q to quit.

RUN THIS

```
$ less notes.txt
```

Kali class notes

Lab tip less is better than cat for long files.

grep

Search inside text

COMMAND grep

> Finds lines that contain a word.

> Great for notes and logs.

> Use -i for case-insensitive search.

RUN THIS

```
$ grep -i "pwd" notes.txt
```

Command: pwd

Lab tip grep answers: where is this word inside the file?

find

Search for files and folders

COMMAND `find`

> Looks for files by name or type.

> Can search inside many folders.

> Start with simple name searches.

RUN THIS

```
$ find . -name "*.txt"
```

```
./notes.txt
```

Lab tip The dot . means search from the current folder.

chmod

Change file permissions

COMMAND **chmod**

- Controls who can read, write, or execute.
- Use +x to make a script executable.
- Permissions protect files.

RUN THIS

```
$ touch script.sh
$ chmod +x script.sh
$ ls -l script.sh
```

```
-rwxr-xr-x 1 student student script.sh
```

Lab tip x means execute, which allows a script to run.

chown

Change file owner

COMMAND chown

> Changes who owns a file.

> Often needs sudo.

> Useful when permission errors happen.

RUN THIS

```
$ sudo chown student:student notes.txt
```

Lab tip Owner and permissions work together.

sudo

Run as administrator

COMMAND sudo

- Runs one command with admin power.
- It may ask for your password.
- Use it only when needed.

RUN THIS

```
$ sudo apt update
```

```
[sudo] password for student:
```

Careful sudo can change the system. Read the command before pressing Enter.

apt update

Refresh package lists

COMMAND apt update

- Checks what software versions are available.
- Does not install updates by itself.
- Run before installing packages.

RUN THIS

```
$ sudo apt update
```

```
Reading package lists... Done
```

Lab tip Think of update as refreshing the software catalog.

apt install

Install software

COMMAND apt install

- Installs a package from repositories.
- Needs sudo on most systems.
- Use exact package names.

RUN THIS

```
$ sudo apt install tree
```

```
Do you want to continue? [Y/n]
```

Lab tip For class, install only tools your instructor asks for.

ip

Show network information

COMMAND `ip`

> Shows network interfaces and IP addresses.

> Use ip a for address information.

> Modern replacement for ifconfig.

RUN THIS

```
$ ip a
```

```
2: eth0: ...
    inet 192.168.56.101/24
```

Lab tip Your IP address identifies your machine on the network.

ping

Test if a host responds

COMMAND ping

> Sends small network test packets.

> Good for checking connectivity.

> Stop with Ctrl+C.

RUN THIS

```
$ ping -c 4 8.8.8.8
```

4 packets transmitted, 4 received

Lab tip Use ping to ask: can I reach this address?

curl

Fetch web content

COMMAND curl

> Downloads or prints data from a URL.

> Common for APIs and websites.

> Use -I to show headers only.

RUN THIS

```
$ curl -I https://example.com
```

```
HTTP/2 200
content-type: text/html
```

Lab tip curl is useful when testing web services from the terminal.

wget

Download a file

COMMAND wget

> Downloads a file from a URL.

> Simple for saving files.

> Shows progress while downloading.

RUN THIS

```
$ wget https://example.com/index.html
```

```
index.html saved
```

Lab tip Use wget when you want a file saved to disk.

tar

Create or extract archives

COMMAND tar

> Packs many files into one archive.

> Common on Linux systems.

> Use -czf to create .tar.gz files.

RUN THIS

```
$ tar -czf kali-lab.tar.gz kali-lab
$ ls
```

```
kali-lab  kali-lab.tar.gz
```

Lab tip — tar is useful before sharing or backing up a lab folder.

nmap

Scan your own lab machine

COMMAND **nmap**

> Shows open network services.

> Use only on systems you own or have permission to test.

> Start with a simple safe scan.

RUN THIS

```
$ nmap 127.0.0.1
```

```
PORT   STATE SERVICE
22/tcp open  ssh
```

Ethics For freshmen: scan localhost or a teacher-approved lab target only.